INCH-POUND
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FEDERAL SPECIFICATION

FELT SHEET: CLOTH, FELT, WOOL, PRESSED

This specification is approved for use by all Departments and Agencies of the Department of the Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for pressed, mechanical-roll, roll-sheet, and sheet wool felt cloths used on several applications (see 6.6).

1.2. Classification. The felt cloths will be available in the following types and classes as specified (see 6.2).

1.2.1 Types and classes.

Type I - Mechanical- roll felt

Type II - Sheet felt
Class 1 - 100% wool, U.S. Standard 62's
Class 2 - 100% wool, U.S. Standard 58's
Class 3 - 75% wool U.S. Standard 56's, 25% wool U.S. Standard 48's
Class 4 - 60% wool U.S. Standard 50's, 40% wool U.S. Standard 44's

Type III - Roll-felt (apparel and decorative)
Type IV - Needle-punched roll felt

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2312 - Felt, Hair or Wool: Mildew Resistant and Moisture Resistant Treatment for
MIL-DTL-43665 - Cloth Wool: Mothproofing Treatment of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Wool Products Labeling Act

(Copies of this document are available online at https://www.ftc.gov.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

United States (U.S.) Standard for Grades of Wool

(Copies of this document are available online at https://www.ams.usda.gov.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9	- Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM20A	- Test Method for Fiber Analysis: Quantitative
AATCC TM81	- Test Method for pH of the Water-Extract from Wet Processed Textiles

(Copies of these documents are available on line at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1777	- Standard Test Method for Thickness of Textile Materials
ASTM D3776/D3776M	- Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5430	- Standard Test Methods for Visually Inspecting and Grading Fabrics

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Standard sample. The finished felt shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Wool. The wool fibers shall be fleeced, pulled wool, wool noils, reprocessed wool, reused wool, or a combination thereof. Any wool shall be acceptable for Type I and Type IV. The grade of the wool shall conform to or be finer than the U.S. standards specified in 3.4.1.1 for Type II and in Table III for Type III.

3.4.1.1 Type II felt. The grades of wool for Type II felts shall be as follows:

Class 1. White 100% wool, U.S. Standard 62's or finer, processed to meet the requirements of this document.

Class 2. White 100% wool, U.S. Standard 58's or finer, processed to meet the requirements of this document.

Class 3. Composed of 75% wool U.S. Standard 56's, 25% wool U.S. Standard 48's or finer, processed to meet the requirements of this document.

Class 4. Composed of 60% wool U.S. Standard 50's, 40% wool U.S. Standard 44's or finer, processed to meet the requirements of this document.

3.5 Classification codes. The felts are classified by a code letter preceded by a numeral to indicate consistency in terms of density (unit weight in pounds per square yard of 1-inch nominal thickness) and followed by a second numeral to indicate the difference in class based on specification requirements. The letter "R" in type I refers to roll felt; the letter "S" in Type II refers to sheet felt; the letter "A" in Type III refers to apparel and decorative felts and the letter "N" in Type IV refers to needle-punched felt. For more specific information and on the numeral notations after the letter for each type see 6.6.

3.6 Color.

3.6.1 Color. The color of the finished felts shall be as specified in the contract or purchase order (see 6.2).

3.6.2 Visual shade matching (all classes). The color and appearance of the dyed felt shall match the standard sample (see 6.3) when tested as specified in 4.5.

3.6.3 Colorfastness. The colorfastness of the finished felts shall be as specified in the contract or purchase order (see 6.2).

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3.7 Physical requirements.

3.7.1 Type I mechanical-roll felt. The Type I felt shall conform to the physical requirements listed below in Table I, when tested in accordance with the test methods specified in 4.5.

TABLE I. Physical requirements for Type I felts.

Classification number	Color	Width, inches (min.)	Percent wool fiber content 1/ (min.)	Breaking strength, lbs./in.^{2} 2/ (min.)	Splitting resistance, lbs./2-inches of width 2/ (min.)	Thickness, inches	Weight, lbs./yd^2
18R1	White	60 or 72	95	600	35	0.125 ±0.012	2.25 ±0.10
						0.188 ±0.013	3.38 ±0.15
						0.250 ±0.014	4.50 ±0.20
						0.313 ±0.015	5.63 ±0.25
						0.375 ±0.016	6.75 ±0.30
						0.500 ±0.019	9.0 ±0.40
16R1X	White	60 or 72	95	500	-	0.047 ±0.007	0.75 ±0.04
						0.062 ±0.007	0.98 ±0.03
						0.078 ±0.007	1.20 ±0.04
16R3X	Gray	60 or 72	92	300		0.094 ±0.007	1.42 ±0.04
16R1	White	60 or 72	95	500	33	0.063 ±0.013	1.0 ±0.05
						0.125 ±0.012	2.0 ±0.10
						0.188 ±0.013	3.0 ±0.15
						0.250 ±0.014	4.0 ±0.20
						0.313 ±0.015	5.0 ±0.25
						0.375 ±0.016	6.0 ±0.30
						0.500 ±0.019	8.0 ±0.40
16R2	Any, except gray or black	60 or 72	90	500	28	0.625 ±0.022	10.0 ±0.50
						0.750 ±0.025	12.0 ±0.60
						0.875 ±0.028	14.0 ±0.70
						1.000 ±0.031	16.0 ±0.80
16R3	Gray	60 or 72	85	400	22	0.125 ±0.012	2.0 ±0.10
						0.188 ±0.013	3.0 ±0.20
						0.250 ±0.014	4.0 ±0.24
						0.313 ±0.015	5.0 ±0.25
						0.375 ±0.016	6.0 ±0.30
						0.500 ±0.019	8.0 ±0.40
						0.625 ±0.022	10.0 ±0.50
						0.750 ±0.025	12.0 ±0.60
						0.880 ±0.028	14.0 ±0.80
						1.00 ±0.031	16.0 ±0.80

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TABLE I. Physical requirements for Type I felts. - Continued

Classification number	Color	Width, inches (min.)	Percent wool fiber content 1/ (min.)	Breaking strength, lbs./in.^{2} 2/ (min.)	Splitting resistance, lbs./2-inches of width 2/ (min.)	Thickness, inches	Weight, lbs./yd^2
12RX3	Gray or Black	60 or 72	75	200	-	0.063 ±0.013	0.75 ±0.04
						0.094 ±0.012	1.12 ±0.035
12R1	White	60 or 72	95	400	18	0.125 ±0.014	1.53 ±0.08
						0.188 ±0.016	2.29 ±0.12
						0.250 ±0.018	3.06 ±0.16
						0.313 ±0.020	3.82 ±0.20
						0.375 ±0.022	4.59 ±0.24
						0.500 ±0.052	6.3 ±0.50
						0.625 ±0.060	7.65 ±0.40
12R2	Gray	60 or 72	87	275	16	0.750 ±0.068	9.18 ±0.48
						0.875 ±0.076	10.71 ±0.56
12R3	Gray	60 or 72	80	250	12	1.00 ±0.084	12.24 ±0.64
9R1	White	72	95	225	8	0.063 ±0.013	0.6 ±0.05
						0.125 ±0.012	1.06 ±0.08
						0.188 ±0.023	1.53 ±0.12
						0.250 ±0.018	2.12 ±0.16
9R2	Gray	72	87	200	6	0.313 ±0.029	2.62 ±0.20
9R3	Gray	72	85	100	3	0.375 ±0.032	3.18 ±0.24
9R4 3/	Gray	72	75	75	2	0.500 ±0.038	4.24 ±0.32
9R5	Gray	72	55	75	2	0.625 ±0.044	5.30 ±0.40
						0.750 ±0.050	6.36 ±0.48
						0.880 ±0.060	7.42 ±0.56
					4/	1.000 ±0.062	8.48 ±0.64
						2.000 ±0.160	16.9 ±1.28
8R5	Gray	72 or 36	45	-	-	0.125 ±0.040	0.84 ±0.03
						0.250 ±0.056	1.80 ±0.18
						0.375 ±0.072	2.70 ±0.27
						0.500 ±0.090	3.60 ±0.36
						0.750 ±0.120	5.40 ±0.54
						1.000 ±0.150	7.20 ±0.72

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1/ The wool fiber content shall be based on the original dry weight of the specimen when tested as specified in 4.5.
2/..The requirement shall be applicable to both lengthwise and crosswise directions separately.
3/ To achieve 2-inch thickness, bond two 1-inch thick layers together placing double-faced film between two layers of felt and applying pressure without heating the material using neoprene or acrylic adhesive.
4/ Sheet size shall be specified minimum.

3.7.2 Type II sheet-felt. The Type II sheet felt shall conform to the physical requirements listed below in Table II, when tested in accordance with the test methods specified in 4.5.

TABLE II. Physical requirements for Type II sheet felts.

Classification number	Color 1/	Percent wool fiber content 2/ (min.)	Breaking strength, lbs./in.^{2} 3/ (min.)	Splitting resistance, lbs./2-inches of width 3/ (min.)	Water thickness swell, percent (max.)	Thickness, inches 4/	Weight, lbs./yd^2
12-S1	White	95	400	18	20	0.125 ±0.020	3.00 ±0.20
						0.375 ±0.025	4.50 ±0.30
						0.500 ±0.030	6.00 ±0.40
						0.625 ±0.035	7.50 ±0.45
						0.750 ±0.040	9.00 ±0.50
12-S2	White	95	300	16	25	0.880 ±0.046	10.50 ±0.55
						1.000 ±0.051	12.00 ±0.60
12-S3	White	95	300	12	30	1.250 ±0.062	15.00 ±0.70
						1.500 ±0.072	18.00 ±0.80
12-S4	White	95	300	10	30	1.750 ±0.083	21.00±0.90
						2.000 ±0.094	24.00 ±1.0
						2.500 ±0.115	30.00 ±1.05
						3.000 ±0.136	36.00 ±1.10
16-S1	White	95	500	32	25	0.125 ±0.012	2.00 ±0.10
						0.188 ±0.012	3.00 ±0.15
						0.250 ±0.017	4.00 ±0.30
						0.375 ±0.020	6.00 ±0.40
						0.500 ±0.022	8.00 ±0.50
16-S2	White	95	400	28	30	0.625 ±0.026	10.00 ±0.60
						0.750 ±0.029	12.00 ±0.70
16-S3	White	95	400	22	35	0.880 ±0.032	14.00 ±0.75
						1.000 ±0.036	16.00 ±0.80

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TABLE II. Physical requirements for Type II sheet felts. - Continued

Classification number	Color 1/	Percent wool fiber content 2/ (min.)	Breaking strength, lbs./in.^{2} 3/ (min.)	Splitting resistance, lbs./2-inches of width 3/ (min.)	Water thickness swell, percent (max.)	Thickness, inches 4/	Weight, lbs./yd^2
16-S4	White	95	300	20	35	1.250 ±0.042	20.00 ±0.90
						1.500 ±0.049	24.00 ±1.00
						1.750 ±0.056	28.00 ±1.10
						2.000 ±0.063	32.00 ±1.20
						2.500 ±0.076	40.00 ±1.25
						3.000 ±0.09	48.00 ±1.30
20-S1	White	95	600	44	30	0.125 ±0.012	2.50 ±0.30
						0.188 ±0.012	2.75 ±0.35
						0.250 ±0.014	5.00 ±0.40
						0.375 ±0.016	7.50 ±0.50
						0.500 ±0.019	10.00 ±0.60
20-S2	White	95	500	40	40	0.625 ±0.022	12.50±0.70
						0.750 ±0.019	15.00±0.80
20-S3	White	95	400	36	50	0.880 ±0.027	17.50 ±0.90
						1.000 ±0.030	20.00 ±1.00
20-S4	White	95	300	32	50	1.250 ±0.036	25.00 ±1.10
						1.50 ±0.041	30.00 ±1.20
						1.750 ±0.047	35.00 ±1.30
						2.000 ±0.053	40.00 ±1.40
						2.500 ±0.064	50.00 ±1.45
						3.000 ±0.075	60.00 ±1.50
26-S1	White	95	600	48	40	0.125 ±0.010	3.25 ±0.40
						0.188 ±0.010	4.90 ±0.50
						0.250 ±0.013	6.50 ±0.60
						0.375 ±0.013	9.75 ±0.80
						0.500 ±0.016	13.00 ±1.0
26-S2	White	95	500	46	50	0.625 ±0.018	16.25 ±1.10
						0.750 ±0.020	19.50 ±1.20
						0.880 ±0.022	22.75 ±1.30
26-S3	White	95	400	40	60	1.000 ±0.025	26.00 ±1.40
						1.250 ±0.025	32.50 ±1.50

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TABLE II. Physical requirements for Type II sheet felts. - Continued

Classification number	Color 1/	Percent wool fiber content 2/ (min.)	Breaking strength, lbs./in.^{2} 3/ (min.)	Splitting resistance, lbs./2-inches of width 3/ (min.)	Water thickness swell, percent (max.)	Thickness, inches 4/	Weight, lbs./yd^2
26-S4	White	95	400	30	60	1.500 ±0.033	39.00 ±1.60
						1.750 ±0.038	45.50 ±1.70
						2.000 ±0.042	52.00 ±1.80
32-S1	White	95	600	50	50	0.125 ±0.007	4.00 ±0.50
						0.188 ±0.007	6.00 ±0.75
						0.250 ±0.007	8.00 ±1.00
						0.375 ±0.009	12.00 ±1.10
						0.500 ±0.011	16.00 ±1.20
32-S2	White	95	500	48	60	0.625 ±0.013	20.00 ±1.30
						0.750 ±0.015	24.00 ±1.40
32-S3	White	95	400	46	70	0.880 ±0.017	28.00 ±1.50
						1.000 ±0.019	32.00 ±1.60
32-S4	White	95	400	40	70	1.250 ±0.022	40.00 ±1.70
						1.50 ±0.026	48.00 ±1.80
						1.750 ±0.033	56.00 ±1.90
						2.00 ±0.032	64.00 ±2.00

1/ Other colors are manufactured on order.
2/ The wool fiber content shall be based on the original dry weight of the specimen when tested as specified in 4.5.
3/ The requirement shall be applicable to both lengthwise and crosswise directions separately.
4/ Sheets of 1/16-inch thickness are manufactured on order for classification numbers 26-S and 32-S.

3.7.3 Type III roll-felt. The Type III felt shall conform to the physical requirements listed below in Table III, when tested in accordance with the test methods specified in 4.5.

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TABLE III. Physical requirements for Type III felts.

Classification number	Trade designation	Grade of wool	Color	Percent wool fiber content 1/ (min.)	Width inches (min.)	Breaking strength, lbs. 2/ (min.)	Thickness, inches	Weight
								oz./yd^2
11A2	Coat front	48's	Gray	20	80	-	0.030 ±0.005	3.04 ±0.11
11A1	Coat front	48's	White	20	80	-	0.030 ±0.005	3.04 ±0.11
10A2	Lining	48's	Gray	45	72	8	0.065 ±0.010	6.00 ±0.25
10A1	Lining, shoe tongue	48's	White	35	72	10	0.065 ±0.010	6.00 ±0.25
9A2	Lining	48's	Gray	45	72	10	0.075 ±0.015	7.00 ±0.25
8A1	Undercollar face mask	58's	White & all colors	95	72	30	0.040 ±0.010	8.00 ±0.50
7A1	Chevron facing, cap	62's	White & all colors	95	60	45	0.040 ±0.010	10.0 ±0.75
6A1	Chevron backing, auto flags & face mask	62's	White & all colors	95	72	30	0.063 ±0.010	12.0 ±1.00
								lbs./yd^2
5A1	Orthopedic truss & athletic pad	56's	White	35	72	-	0.125 ±0.04	1.00 ±0.06
							0.250 ±0.051	2.00 ±0.13
						-	0.375 ±0.064	3.00 ±0.25
						-	0.500 ±0.075	4.00 ±0.25
4A1	Surgical	56's	White	50	36	20	0.063 ±0.04	0.53 ±0.03
		56's	White	50	72	45	0.125 ±0.04	1.06 ±0.06
		56's	White	50	72	175	0.375 ±0.04	3.00 ±0.12
3A1	Mid-sole	56's	Gray	95	60 by 40 sheets	-	0.250 ±0.051	7.15 ±0.65

1/ The wool fiber content shall be based on the original dry weight of the specimen when tested as specified in 4.5.
2/ The requirement shall be applicable to both lengthwise and crosswise separately.

3.7.4 Type IV needle-punched roll-felt. The Type IV felt shall conform to the physical and chemical requirements listed below in Table IV, when tested in accordance with the test methods specified in 4.5.

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TABLE IV. Physical requirements for Type IV needle-punched felts.

Classification number	Trade designation	Color	Percent wool fiber content 1/ (min.)	Width inches (min.)	Splitting resistance, lbs./2-inches of width 2/ (min.)	Breaking strength, lbs./in² 2/ (min.)	Thickness, inches	Weight, lbs./yd²
12NS 3/	Needle punched	Gray	45	72	14	525	0.500 ±0.25	5.80-6.80

1/ The wool fiber content shall be based on the original dry weight of the specimen when tested as specified in 4.5.
2/ The requirement shall be applicable to both lengthwise and crosswise separately.
3/ The felt shall also contain a minimum of 45 percent wool, 45 percent polyester and a maximum of 5 percent polypropylene and 5 percent unspecified fibers.

3.7.5 Width. Types I, II, III and IV felts shall be trimmed and the minimum width after trimming shall be as specified in Tables I, III and IV.

3.7.6 Ammunition felt. When ammunition felt is specified (see 6.2), the requirements cited in 3.7.6.1 and 3.7.6.2 shall apply.

3.7.6.1 pH and organic acidity. The pH of water extract of the ammunition felt shall be 7.0 (± 1.0) and the acidity of the extract calculated as acetic acid shall not exceed 0.12 percent when tested as specified in 4.5.

3.7.6.2 Breaking strength. Ammunition felt of 1/16 and 3/32-inch thickness in classification number 12R3X shall a have a minimum tensile strength of 125 lbs. per square inch (p.s.i.) for both lengthwise and crosswise directions when tested as specified in 4.5.

3.7.7 Mildew resistant treatment. When specified (see 6.2), the felt shall be mildew resistant treated in accordance with MIL-PRF-2312 except that the 9R-1 through 9R-5, 2-inch, the complete penetration of the treatment shall not be required and the contractor shall certify the kind of inhibitor which was applied. The physical characteristics of the treated felt shall conform to the requirements specified herein.

3.7.8 Mothproofing treatment. The felts shall be mothproofed treated in accordance with MIL-DTL-43665. The physical characteristics of the treated felt shall conform to the requirements specified herein.

3.8 Wool content identification label. The finished felts shall be labeled or ticketed for wool/fiber content in accordance with the Rules and Regulations Under the Wool Products Labeling Act and the U.S. Standard for Grades of Wool.

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3.9 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the put-up length shall be as follows:

3.9.1 Types I and IV. Types I and IV felt shall be furnished in rolls of the following lengths. Each roll shall contain no more than two pieces, neither of which shall be less than 3-yards in length. When specified (see 6.2), Type I, 3/4 to 2 inch felt may be put up in sheets 72 by 36 inches (minimum).

Thickness (inches)	Length (yards)
5/16 and under	26 to 55
Over 5/16 but not exceeding 1/2	13 to 33
Over 1/2 to 1 inclusive	7 to 17
2	3 to 7

3.9.2 Type II. Type II felt shall be furnished in sheets 36 by 36 inches (± 1/2) inch.

3.9.3 Type III. The Type III felt shall be furnished in rolls of 28 to 80 yards in length for 60, 72, and 80 inch widths, and 24 to 24-1/2 yards in length for 36-inch width. Each roll shall contain no more than two (2) pieces; no piece shall be less than 5-yards in length.

3.10 Toxicity. The finished felts shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.11 Workmanship. The finished felts shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

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4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. The finished roll or felt sheet shall be examined for the defects listed in MIL-STD-655. The defects specified in Table V shall be scored four (4) points for each treated yard or sheet in which they occur when evaluated in accordance with ASTM D5430.

TABLE V. Visual defects.

Cut, hole or tear
Open, breaks, or thin area
Abrasion or rub
Surface nappiness or glossier than adjoining normal areas
Wrinkles and creases
Surface pucker or ripple
Waviness or inability to lay flat
High, thick, or lumpy area
Burrs, shives, seed, or other naturally occurring foreign matter
Spots, stains, or streaks not due to fiber distribution or bleeding
Scalloped, cutout, narrow, jagged or untrimmed edge extending 9 inches of more on a side
Soft area noticeably less firm than adjoining normal areas
Extraneous foreign matter

4.4.1 Wool grade examination. Fleeced or pulled wool shall be visually examined for grade (see 3.4.1.1 and Table III) in scoured form prior to dyeing, picking, or carding by comparison with the applicable United States (U.S.) Standard for Grades of Wool. The examination for grade shall be performed on a composite sample of 1 pound of wool for each 10,000 pounds or fraction thereof of the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot.

4.4.2 Noils. Noils shall be examined prior to dyeing and blending. Sampling shall be as specified in 4.4.1.

4.4.3. Reprocessed and reused wool. Reprocessed and reused wool shall be examined for grade when in fibrous state prior to blending with other fibers. Sampling shall be as specified in 4.4.1.

4.4.4 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll or sheet in the sample shall be examined for defects as specified in MIL-STD-655.

4.4.5 Shade, finish and appearance examination. During the yard-by-yard examination, each roll or sheet in the sample shall be examined for shade variation as specified in MIL-STD-655.

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4.4.6 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-655 and see 3.9.

4.5 End item testing. The felts shall be tested for the characteristics listed in Table VI. The methods of testing as specified wherever applicable and as listed in Table VI shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) sheets or 3-1/2 continuous yards full width of the finished felt for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VI. End item tests.

Characteristics	Requirement	Test Method
Wool content (all types)	Table I, II, III and IV	AATCC 20A (see 6.5)
Polyester content (Type IV) Polypropylene content (Type IV)	Table IV Table IV	AATCC 20A (see 6.5)
Wool grade Type II Type III	 3.4.1.1 Table III	U.S. Standards for Grades of Wool
Visual shade matching	3.6.2	4.6.1

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TABLE VI. End item tests. - Continued

Characteristics	Requirement	Test Method
Weight Type I, lbs./yd² Type II, lbs./yd² Type III (11A2, 11A1, 10A2, 10A1, 9A2, 8A1, 7A1 and 6A1), oz./yd² Type III (5A1, 4A1, 3A1), lbs./yd² Type IV lbs./yd²	 Table I Table II Table III Table III Table IV	 ASTM D3776/D3776M ASTM D3776/D3776M ASTM D3776/D3776M ASTM D3776/D3776M ASTM D3776/D3776M
Thickness, (all types), 1/, inches	Table I, II, III, and IV	ASTM D1777
Breaking strength, (Types I, II and IV), lbs./in² 2/ Lengthwise Crosswise	Table I, II, and IV and 3.7.6.2	ASTM D5034
Breaking strength, (Type III), pounds Lengthwise Crosswise	Table III	ASTM D5034
Splitting resistance 3/, (Type I, II and IV) lbs. per 2-inch width Lengthwise Crosswise	Table I, II and IV	4.6.2
Water thickness swell, (Type II), percent	Table II	4.6.3
Organic acidity	3.7.6.1	AATCC TM81 and 4.6.4
Mildew Resistance	3.7.7	4/
Mothproofing	3.7.8	5/
Toxicity	3.10	4.7

1/ Except that three specimens of not less than 40 square inches each shall be used and five determinations shall be made on each specimen.
2/ ASTM D5034, except ten (10) 2 by 6-inch specimens shall be cute from each test sample, five (5) taken lengthwise and five (5) taken crosswise directions. The jaws of the clamps shall be not less than 2-inches in width. The thickness of each specimen shall be measured and the cross-sectional area of each specimen shall be calculated before testing. The breaking strength shall equal the breaking load divided by the cross-sectional area.
3/ Not required on felt less than 3/16-inch thick.
4/ Treated in accordance with MIL-PRF-2312.
5/ Treated in accordance with MIL-DTL-43665.

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4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloths shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 6.4), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

4.6.2 Splitting resistance. The splitting resistance shall be determined using the apparatus as specified in ASTM D5034. Cut ten 2-inch by 6-inch specimens from each sample unit, five (5) taken with the 6-inch dimension parallel to the length and five (5) perpendicular to the length. Split the conditioned specimens within the middle third of the thickness for a distance of approximately 2-inches from one end only. Clamp one of the lips in each jaw of the testing machine. Start the machine and record the average load in pounds necessary to pull the two sections of the felt wholly apart as the splitting resistance in pounds. Average the test results on the five specimens cut lengthwise and on the five specimens cut crosswise, respectively, and report separately as the splitting resistance in pounds per 2-inches of width. This test is not required for materials less than 3/16 inch thick.

4.6.3 Water thickness swell. Three (3) specimens, each approximately 2 by 2 inches, shall be cut from different portions of the sheet. The thickness of the specimen shall be accurately measured in accordance with ASTM D1777. The specimens shall then be placed on a wire screen having ¼ inch opening and immersed in boiling water for two (2) minutes. A wetting agent shall be used for hard felts which will not wet out thoroughly in boiling water in two (2) minutes. The specimens shall be removed and dried to a constant weight at 221 (± 10) ºF. The percentage of water thickness swell shall be calculated as follows:

$$\text{Thickness swell, percent} = \frac{T2-T1}{T1} \times 100$$

Where T1 = Average thickness of original specimen
T2 = Average thickness of specimen after wetting and drying

4.6.4 Organic acidity. Use the water extract from the pH test (AATCC TM81). Titrate the solution potentiometrically to a pH of 8.3 using a glass electrode and 0.1N sodium hydroxide solution. Make a blank determination and calculate the percent organic acidity of the water extract as acetic acid as follows:

$$\text{Percent organic acidity} = \frac{6.005\,(V-v)N}{W}$$

Where: V = milliliters of sodium hydroxide solution used to titrate that water extract
v = milliliters of sodium hydroxide solution used to titrate the blank
N = normality of the sodium hydroxide solution
W = weight of the specimen (grams)

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4.7 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.10) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Tables VII, VIII, IX, and X shows the suggested applications for the various classifications of Types I, II, III and IV felts respectively.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.2)
e. Color required (see 3.6 and 6.3)
f. Width of felt required (see 3.7.5)
g. Length required if other than specified (see 3.9)
h. When toxicity testing is required (see 3.10, 4.7)
i. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
j. Inspection conditions (see 4.3.1)
k. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

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6.4 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5)º angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.6 Felts classifications and suggested use.

6.6.1 Type I felt. The Type I felts are classified by the code letter "R" preceded by a numeral to indicate consistency in terms of density (unit weight in pounds per square yard of 1-inch nominal thickness) and followed by a second numeral to indicate the difference in class based on specification requirements which are listed in Table I and as follows:

Classification	Class	Trade designation
8R	5	Soft pad
9R	1,2,3,4,5	Firm pad
12R	1,2,3 3X	Extra firm pad Lining
16R	1,2,3 1X,3X	Backcheck Ball bearing
18R	1	Laundry

Note: The letter "X" indicates felt less than 1/8 inch in thickness.

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TABLE VII. Suggested service application for Type I felts.

Classification number	Suggested service application
18R1	For uses where a hard, high-grade felt possessing long wearing properties is desired.
16R1	For oil retention in installations where the felt is not compressed, for feeding low viscosity or light oil, and where unusual strength and hardness are required. Washers; bushings; wicks; ink rolls and pads; door bumpers; polishing blocks; wheels and pads; grommets; window channels; resilient mountings, anti-vibration and dampening pads; and parts where wear and resistance to abrasion are required are typical uses.
16R2	For vibration mounting, oil and grease shields, and the same general purposes as 16R1 where a felt of slightly lower quality is satisfactory.
16R3	For ammunition components. For aircraft applications, between rocker arm covers of engines, ring cowlings radio cushion strips, retaining and feeding oil under difficult conditions, washers and bushings.
12R1 12R2 12R3	For dust shields, wipers, grease retainer washers, wicks, vibration mountings, and uses where a resilient felt is required.
9R1 9R2 9R3	For grease and oil retention where the felt is confined and compressed in assembly. Also recommended for dust shields.
9R4	For sound deadening, chassis strips, spacers, dust shields, pedal pads, dash liners, and for mechanical purposes where abrasion and wear are not important factors.
8R5	For packing or padding when held in place between other materials. This grade should not be used for mechanical purposes.
16R1X	For ball and roller bearing oil retainer washers and small dust excluding washers. Also for mechanical purposes where an accurate, thin, smooth high-grade felt is required.
16R3X	For the same general uses as 16R1X but in installations where tolerances and length of life are not as important. Also for thin cut parts such as gaskets and liners.
12R3X	For antisqueak strips and for lining when cemented to fiberboard or metal panels

6.6.2 Type II felts. The Type II felts are classified by the code letter “S” preceded by a numeral to indicate consistency in terms of density (unit weight in pounds per square yard of 1-inch nominal thickness) and followed by a second numeral to indicate the difference in class based on specification requirements which are listed in Table II and as follows:

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Classification	Class	Trade designation
32S	1 to 4	Extra hard
26S	1 to 4	Hard
20S	1 to 4	Medium
16S	1 to 4	Soft
12S	1 to 4	Extra soft

TABLE VIII. Suggested service application for Type II felts.

Classification number	Qualities	Service application
32S Extra hard	32S1 Fine 32S2 Medium fine 32S3 Medium coarse 32S4 Coarse	For use as extra hard-density polishing wheels and buffs in dental, jewelry, glass, and lapidary polishing: also hard washers, bumpers, and casters.
26S Hard	26S1 Fine 26S2 Medium fine 26S3 Medium coarse 26S4 Coarse	For use as hard-density polishing wheels for glass sheet, glassware, ophthalmic lens polishing, metal and metallographic polishing, wood polishing, and furniture rubbing, also for block cutters, print rolls, cash carrier heads, points for marking pens, casters, boot and shoe soles, artificial limbs.
20S Medium	20S1 Fine 20S2 Medium fine 20S3 Medium coarse 20S4 Coarse	For use as medium hard-density polishing wheels and buffs for lens polishing, mirrors and glass, marble and granite; also for fluid transfer rolls, ink rolls, securing, furniture rubbing, rough metal polishing, metal wiping, drilled wicks, bearing seal washers, stamp pads, cushioning under sandpaper.
16S Soft	16S1 Fine 16S2 Medium fine 16S3 Medium coarse 16S4 Coarse	For use as medium-density polishing wheels and buffs for precious metal and plastic polishing, rough optical polishing, metal wiping, drum beaters; also drilled wicks, bearing seals, shoe rolls (shank), fluid transfer rolls, oil and fluid wicks, grease and oil retaining washers, ink rollers, vibration and shock mountings, bumpers, plugs, glass channels.
12S Extra soft	12S1 Fine 12S2 Medium fine 12S3 Medium coarse 12S4 Coarse	For use as soft-density polishing wheels and buffs for polishing plastic, polishing and wiping brass; also for piano wedge, surgical pads, punched wicks, dampeners, absorbent pads, oil and fluid retainers, fluid transfer rolls, bearing seals, washers, wicks, shim and spacer pads, shoe insoles, dust shields, anti-vibration pads.

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6.6.3 Type III felts. The Type III felts covered by this specification are classified by the code letter "A" preceded by a numeral to indicate difference in commercial designation and followed by a second numeral to indicate the class based on specification requirements which are listed in Table III and as follows:

Classification	Class	Trade designation
11A	1, 2	Coat fronts
10A	1, 2	Lining; shoe tongue
9A	2	Lining
8A	1	Undercollar, face mask
7A	1	Chevron facing, cap
6A	1	Chevron backing, auto flats and face mask
5A	1	Orthopedic truss and athletic padding
4A	1	Surgical
3A	1	Midsole

TABLE IX. Suggested service application for Type III felts.

Trade designation	Suggested service application
Coat front felt	For use in front of coats to give required fullness and drape to outer fabrics
Lining; lining, shoe tongue felt	For use as a lining on inside of the shoe tongue, generally white in color. Backing for household objects to prevent scratching or marking furniture
Lining felt	For use as a lining material in outerwear garments
Chevron, face ask, hat body, undercollar, face mask felt	For use in garment decoration, background, for embroidered designed hat bodies, undercollar cloth, military insignia
Orthopedic truss and athletic padding	For use as padding on orthopedic and truss and appliances, athletic equipment padding
Surgical	This type of felt can be pulled down in layers to graduated thicknesses needed for medical requirements
Midsole	For use in footwear as lining or inserts

6.6.3 Type IV felts. The Type IV felts are classified by the code letter "N" preceded by a numeral to indicate consistency in terms of density (unit weight in pounds per square yard of 1-inch nominal thickness) and followed by a second numeral to indicate the difference in class based on specification requirements which are listed in Table IV and as follows:

Classification	Class	Trade designation
12N5	5	Needle-punched

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TABLE X. Suggested service application for Type IV felts.

Classification number	Suggested service application
12N5	For dust shields, wipers, grease retainer washers, vibration mountings, and uses where a resilient felt is required

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Subject term (key word) listing.

Coat fronts
Mildew resistant
Mothproofing
Needle-punch
Roll

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - NU
Air Force - 11

Review activities:
Army - AR, MD
Navy - MC, OS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FAS

Preparing Activity:
DLA-CT

(Project: 8305-2021-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

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